Distributed Data Engineering Pipeline for Tech Job Market Analysis

Your Name

1 Project Overview

This project implements a scalable, end-to-end data engineering pipeline designed to collect, process, and analyze technology job market data from multiple online sources. The system follows a modern data lake architecture (Bronze-Silver-Gold) and leverages distributed data processing to handle large volumes of semi-structured data.

The primary objective is to transform raw web-scraped job postings into clean, structured, and analytics-ready datasets that provide insights into job demand, salary trends, and required technical skills in the tech industry.

2 Objectives

The key goals of this project are:

- Build a production-style data pipeline using real-world data
- Implement web scraping for automated data ingestion
- Apply a layered data architecture (Bronze, Silver, Gold)
- Perform distributed data processing using PySpark
- Orchestrate workflows using Apache Airflow
- Ensure data quality and reproducibility
- Generate business-oriented datasets for analytics

3 Data Sources

The pipeline collects job postings from multiple online platforms such as:

- Job boards (e.g., Indeed, LinkedIn, Glassdoor)
- Company career pages

Each job posting typically includes:

- Job title
- Company name
- Location
- Salary (if available)

- Job description
- Required skills
- Posting date

4 System Architecture

The pipeline follows a layered data lake architecture:

4.1 Bronze Layer (Raw Data)

The Bronze layer stores raw, unprocessed data collected from web scraping. Data is stored in JSON or Parquet format with minimal transformation.

Additional metadata fields include:

- ingestion$_{t}imestamp$
- source$_{n}ame$
- scrape$_{d}ate$
- run$_{i}d$

4.2 Silver Layer (Cleaned Data)

The Silver layer applies data cleaning and standardization:

- Schema enforcement and type casting
- Deduplication of job postings
- Parsing salary ranges
- Standardizing locations
- Extracting skills from text
- Handling missing and inconsistent values

The output consists of reliable, structured datasets ready for analysis.

4.3 Gold Layer (Business Data)

The Gold layer contains aggregated, business-oriented datasets such as:

- Job postings per city and country
- Average salary by role
- Most in-demand skills
- Job trends over time
- Remote vs on-site distribution

These datasets are optimized for analytics and can be consumed by BI tools.

5 Technology Stack

- Programming Language: Python
- Web Scraping: BeautifulSoup / Scrapy
- Distributed Processing: PySpark
- Workflow Orchestration: Apache Airflow
- Storage: Parquet (data lake)
- Containerization: Docker
- Database (optional): PostgreSQL

6 Pipeline Workflow

The pipeline consists of the following stages:

1. **Data Ingestion**: Scrape job postings and store raw data in the Bronze layer
2. **Validation**: Perform basic checks on data completeness and structure
3. **Transformation (Silver)**: Clean, normalize, and enrich data using PySpark
4. **Aggregation (Gold)**: Generate analytical datasets and KPIs
5. **Export**: Save outputs in Parquet and CSV formats

7 Orchestration

Apache Airflow is used to manage and automate the pipeline. The DAG includes:

- Scraping tasks
- Data validation
- Spark transformation jobs
- Aggregation tasks
- Data quality checks

The workflow supports scheduling, retries, and logging.

8 Data Quality and Reliability

To ensure data reliability, the pipeline includes:

- Schema validation
- Null value checks
- Duplicate detection
- Consistency checks across sources

9 Scalability Considerations

The pipeline is designed for scalability:

- Distributed processing using Spark
- Partitioned storage (by date and source)
- Modular architecture for easy extension
- Support for incremental ingestion

10 Use Cases

This project enables:

- Analysis of job market trends
- Identification of high-demand skills
- Salary benchmarking
- Geographic distribution of tech jobs

11 Future Improvements

Potential enhancements include:

- Integration with cloud platforms (AWS, GCP)
- Real-time streaming (Kafka)
- Advanced NLP for skill extraction
- CI/CD for pipeline deployment

12 Conclusion

This project demonstrates the design and implementation of a real-world data engineering system. It highlights key concepts such as distributed processing, workflow orchestration, data quality management, and scalable architecture.